01_stn_setup

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos

Everyone talks about ACID, but nobody talks abaout “I”, whish must be considered to avoid affecting the database´s operating performance.

```
USE master;
GO

IF DB_ID('STN_Lab') IS NOT NULL
BEGIN
    ALTER DATABASE STN_Lab SET SINGLE_USER WITH ROLLBACK IMME
DIATE;
    DROP DATABASE STN_Lab;
END
GO

CREATE DATABASE STN_Lab;
GO

USE STN_Lab;
GO

-- ============================================================
==
-- TABLAS CON CLUSTERED INDEX DESDE EL INICIO
-- ============================================================
==
CREATE TABLE dbo.TipoImpuesto (
    TipoImpuestoID  SMALLINT        NOT NULL,
```

```
    Codigo          VARCHAR(10)     NOT NULL,
    Descripcion     VARCHAR(100)    NOT NULL,
    TasaBase        DECIMAL(8,4)    NOT NULL,
    CONSTRAINT PK_TipoImpuesto PRIMARY KEY CLUSTERED (TipoImp
uestoID)
);
GO

CREATE TABLE dbo.Contribuyente (
    ContribuyenteID INT             NOT NULL,
    NIT             VARCHAR(20)     NOT NULL,
    RazonSocial     VARCHAR(200)    NOT NULL,
    TipoPersona     CHAR(1)         NOT NULL,
    DepartamentoID  SMALLINT        NOT NULL,
    FechaRegistro   DATE            NOT NULL,
    Estado          CHAR(1)         NOT NULL,
    CONSTRAINT PK_Contribuyente PRIMARY KEY CLUSTERED (Contri
buyenteID)
);
GO

CREATE TABLE dbo.Declaracion (
    DeclaracionID       INT             NOT NULL,
    ContribuyenteID     INT             NOT NULL,
    TipoImpuestoID      SMALLINT        NOT NULL,
    PeriodoFiscal       CHAR(7)         NOT NULL,
    FechaPresentacion   DATETIME2(3)    NOT NULL,
    MontoDeclarado      DECIMAL(18,2)   NOT NULL,
    MontoImpuesto       DECIMAL(18,2)   NOT NULL,
    Estado              CHAR(2)         NOT NULL,
    UsuarioRegistro     VARCHAR(50)     NOT NULL,
    FechaCreacion       DATETIME2(3)    NOT NULL DEFAULT SYSD
ATETIME(),
    CONSTRAINT PK_Declaracion PRIMARY KEY CLUSTERED (Declarac
ionID)
);
```

```
GO

CREATE TABLE dbo.Pago (
    PagoID              BIGINT          NOT NULL,
    DeclaracionID       INT             NOT NULL,
    ContribuyenteID     INT             NOT NULL,
    FechaPago           DATETIME2(3)    NOT NULL,
    MontoPagado         DECIMAL(18,2)   NOT NULL,
    MediosPago          VARCHAR(20)     NOT NULL,
    NumeroReferencia    VARCHAR(50)     NOT NULL,
    BancoID             SMALLINT        NOT NULL,
    CONSTRAINT PK_Pago PRIMARY KEY CLUSTERED (PagoID)
);
GO

CREATE TABLE dbo.AuditoriaFiscal (
    AuditoriaID             INT             NOT NULL,
    ContribuyenteID         INT             NOT NULL,
    DeclaracionID           INT             NOT NULL,
    FechaInicio             DATE            NOT NULL,
    FechaCierre             DATE            NULL,
    AuditorID               INT             NOT NULL,
    DiferenciaDeterminada   DECIMAL(18,2)   NOT NULL DEFAULT
0,
    Estado                  VARCHAR(20)     NOT NULL,
    CONSTRAINT PK_Auditoria PRIMARY KEY CLUSTERED (AuditoriaI
D)
);
GO

-- ============================================================
==
-- DATOS
-- ============================================================
==
INSERT INTO dbo.TipoImpuesto VALUES
```

```sql
(1, 'ISR',    'Impuesto Sobre la Renta',         0.25),
(2, 'IVA',    'Impuesto al Valor Agregado',       0.12),
(3, 'IUSI',   'Impuesto Único Sobre Inmuebles',   0.009),
(4, 'ISO',    'Impuesto de Solidaridad',          0.01),
(5, 'IETAP',  'Impuesto Específico sobre Tabaco', 0.75),
(6, 'IPRIMA', 'Impuesto sobre Primas de Seguros', 0.03);
GO

;WITH
L0 AS (SELECT 1 AS c UNION ALL SELECT 1),
L1 AS (SELECT 1 AS c FROM L0 a, L0 b),
L2 AS (SELECT 1 AS c FROM L1 a, L1 b),
L3 AS (SELECT 1 AS c FROM L2 a, L2 b),
L4 AS (SELECT 1 AS c FROM L3 a, L3 b),
L5 AS (SELECT 1 AS c FROM L4 a, L4 b),
Nums AS (SELECT TOP 200000 ROW_NUMBER() OVER (ORDER BY (SELEC
T NULL)) AS n FROM L5)
INSERT INTO dbo.Contribuyente
SELECT
    n,
    CAST(n * 7 + 1000000 AS VARCHAR(20)),
    'CONTRIBUYENTE ' + CAST(n AS VARCHAR(10)),
    CASE WHEN n % 3 = 0 THEN 'J' ELSE 'I' END,
    CAST((n % 22) + 1 AS SMALLINT),
    DATEADD(DAY, -(n % 3650), '2024-01-01'),
    CASE WHEN n % 20 = 0 THEN 'I'
         WHEN n % 50 = 0 THEN 'S'
         ELSE 'A' END
FROM Nums;
GO

;WITH
L0 AS (SELECT 1 AS c UNION ALL SELECT 1),
L1 AS (SELECT 1 AS c FROM L0 a, L0 b),
L2 AS (SELECT 1 AS c FROM L1 a, L1 b),
L3 AS (SELECT 1 AS c FROM L2 a, L2 b),
```

```
L4 AS (SELECT 1 AS c FROM L3 a, L3 b),
L5 AS (SELECT 1 AS c FROM L4 a, L4 b),
Nums AS (SELECT TOP 1500000 ROW_NUMBER() OVER (ORDER BY (SELE
CT NULL)) AS n FROM L5)
INSERT INTO dbo.Declaracion
    (DeclaracionID, ContribuyenteID, TipoImpuestoID, PeriodoF
iscal,
     FechaPresentacion, MontoDeclarado, MontoImpuesto, Estad
o, UsuarioRegistro)
SELECT
    n,
    (n % 200000) + 1,
    CAST((n % 6) + 1 AS SMALLINT),
    CAST(2020 + (n % 4) AS VARCHAR(4)) + '-'
        + RIGHT('0' + CAST((n % 12) + 1 AS VARCHAR(2)), 2),
    DATEADD(SECOND, n, '2020-01-01 08:00:00'),
    CAST((n % 500000) + 1000 AS DECIMAL(18,2)),
    CAST(((n % 500000) + 1000) * 0.12 AS DECIMAL(18,2)),
    CASE WHEN n % 15 = 0 THEN 'OM'
         WHEN n % 8  = 0 THEN 'AU'
         ELSE 'PR' END,
    'SISTEMA'
FROM Nums;
GO

;WITH
L0 AS (SELECT 1 AS c UNION ALL SELECT 1),
L1 AS (SELECT 1 AS c FROM L0 a, L0 b),
L2 AS (SELECT 1 AS c FROM L1 a, L1 b),
L3 AS (SELECT 1 AS c FROM L2 a, L2 b),
L4 AS (SELECT 1 AS c FROM L3 a, L3 b),
L5 AS (SELECT 1 AS c FROM L4 a, L4 b),
Nums AS (SELECT TOP 1200000 ROW_NUMBER() OVER (ORDER BY (SELE
CT NULL)) AS n FROM L5)
INSERT INTO dbo.Pago
    (PagoID, DeclaracionID, ContribuyenteID, FechaPago,
```

```
     MontoPagado, MediosPago, NumeroReferencia, BancoID)
SELECT
    n,
    (n % 1500000) + 1,
    (n % 200000) + 1,
    DATEADD(DAY, n % 1460, '2020-01-15'),
    CAST((n % 100000) + 500 AS DECIMAL(18,2)),
    CASE n % 3
        WHEN 0 THEN 'BANCO'
        WHEN 1 THEN 'ONLINE'
        ELSE 'VENTANILLA' END,
    'REF-' + CAST(n * 13 AS VARCHAR(15)),
    CAST((n % 15) + 1 AS SMALLINT)
FROM Nums;
GO

;WITH
L0 AS (SELECT 1 AS c UNION ALL SELECT 1),
L1 AS (SELECT 1 AS c FROM L0 a, L0 b),
L2 AS (SELECT 1 AS c FROM L1 a, L1 b),
L3 AS (SELECT 1 AS c FROM L2 a, L2 b),
L4 AS (SELECT 1 AS c FROM L3 a, L3 b),
L5 AS (SELECT 1 AS c FROM L4 a, L4 b),
Nums AS (SELECT TOP 50000 ROW_NUMBER() OVER (ORDER BY (SELECT
NULL)) AS n FROM L5)
INSERT INTO dbo.AuditoriaFiscal
    (AuditoriaID, ContribuyenteID, DeclaracionID, FechaInici
o,
     FechaCierre, AuditorID, DiferenciaDeterminada, Estado)
SELECT
    n,
    (n % 200000) + 1,
    (n % 1500000) + 1,
    DATEADD(DAY, n % 1000, '2021-01-01'),
    CASE WHEN n % 5 = 0 THEN NULL
         ELSE DATEADD(DAY, (n % 1000) + 90, '2021-01-01') EN
```

```sql
D,
    (n % 50) + 1,
    CAST((n % 250000) AS DECIMAL(18,2)),
    CASE WHEN n % 5 = 0 THEN 'ABIERTA'
         WHEN n % 7 = 0 THEN 'APELADA'
         ELSE 'CERRADA' END
FROM Nums;
GO

-- ==========================================================
==
-- VERIFICACION
-- ==========================================================
==
SELECT 'TipoImpuesto'   AS Tabla, COUNT(*) AS Total FROM dbo.
TipoImpuesto
UNION ALL
SELECT 'Contribuyente',  COUNT(*) FROM dbo.Contribuyente
UNION ALL
SELECT 'Declaracion',    COUNT(*) FROM dbo.Declaracion
UNION ALL
SELECT 'Pago',           COUNT(*) FROM dbo.Pago
UNION ALL
SELECT 'AuditoriaFiscal',COUNT(*) FROM dbo.AuditoriaFiscal;
GO
```

Results | Messages

	Tabla	Total
1	TipoImpuesto	6
2	Contribuyente	200000
3	Declaracion	1500000
4	Pago	1200000
5	AuditoriaFiscal	50000

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (71) | STN_Lab | 00:04:41 | Row: 1, Col: 1 | 5 rows